

FIN301 Exam I - Spring 2026 - Answer Key

Multiple Choice Questions (1-19)

Question 1: Ultimate Goal of Financial Management

Answer: D. Create value for stockholders by maximizing the stock price

The ultimate goal of financial management is to maximize shareholder wealth, which is reflected in the stock price. While reducing costs (A), reducing SG&A (B), and increasing net income (C) can contribute to this goal, they are means to an end rather than the ultimate objective.

Question 2: Accounting Statements

Answer: D. The accounting statement that measures the revenues, expenses, and net income for a certain period of time is called an income statement.

- A is incorrect: Cash flow statement measures cash flows, not revenues/expenses/net income
 - B is incorrect: Balance sheet summarizes assets, liabilities, and equity at a point in time
 - C is incorrect: Balance sheet is a point-in-time statement, not period statement
 - D is correct: Income statement measures revenues, expenses, and net income over a period
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Question 3: Agency Issue (Incorrect Statement)

Answer: A. It refers to the conflict of interest between a principal (CEO) and an agent (stockholders).

This is INCORRECT because:

- The principal is the shareholder (owner), and the agent is the CEO/manager
 - The statement has it reversed - shareholders hire managers (agents) to act on their behalf
 - Options B, C, and D are all correct statements about agency issues
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Question 4: NOT a Good Way to Mitigate Agency Issue

Answer: D. Do not hold a conference call after a firm reports its quarterly earnings so that managers can hide some important information.

- A, B, C are all effective ways to mitigate agency issues
 - D would actually worsen the agency problem by reducing transparency
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Question 5: Sarbanes-Oxley Act (Incorrect Statement)

Answer: D. The SOX came in response to financial scandals in the internet bubble or the dotcom bubble, in the late 1990s.

This is INCORRECT because:

- SOX was passed in 2002 in response to corporate scandals like Enron and WorldCom (2001-2002)
 - It was NOT in response to the dotcom bubble of the late 1990s
 - Options A, B, and C are correct statements about SOX
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Question 6: Compounding Interest

Answer: B. Compounding or compound interest

When Tom earns interest on his prior interest, this is the definition of compound interest. Simple interest (A) would only be earned on the original principal.

Question 7: Present Value Calculation

Answer: D. $PV = \$600 / (1 + 0.07)^6$

Timeline:

Year:	0	1	2	3	4	5	6
	----- ----- ----- ----- ----- -----						
	PV=?						FV=\$600

Present Value Formula: $PV = FV / (1 + r)^n$
 $PV = \$600 / (1.07)^6$

Question 8: Simple Interest Calculation

Answer: C. \$480.00

Timeline:

Date:	1/1/2021	1/1/2022	1/1/2023	1/1/2024	1/1/2025	1/1/2026
	----- ----- ----- ----- -----					
	\$2,400	+\$96	+\$96	+\$96	+\$96	+\$96
	(deposit)	interest	interest	interest	interest	interest

Calculation:

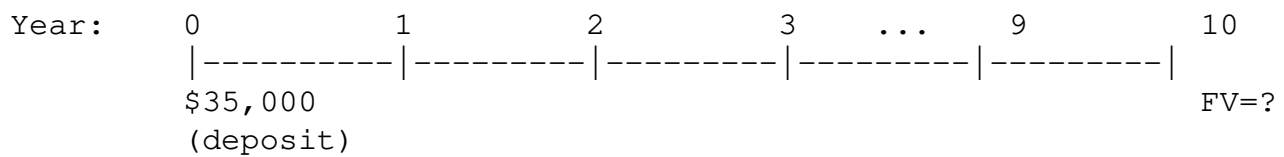
- Simple Interest = Principal $\times$ Rate $\times$ Time
- Simple Interest = $\$2,400 \times 0.04 \times 5 = \480.00

Note: With simple interest, you earn the same \$96 each year (no compounding).

Question 9: Compound Interest - Future Value

Answer: C. \$59,875.06

Timeline:



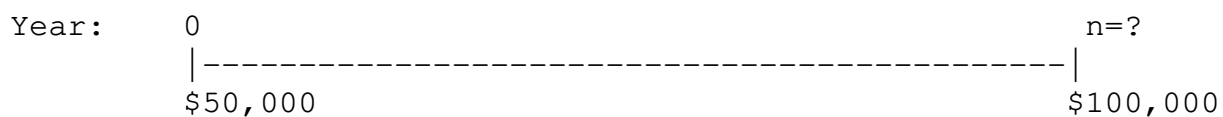
Calculation:

- $FV = PV \times (1 + r)^n$
- $FV = \$35,000 \times (1.055)^{10}$
- $FV = \$35,000 \times 1.7081444$
- $FV = \mathbf{\$59,875.06}$

Question 10: Time to Double (Challenging)

Answer: D. 12.95 years

Timeline:



Calculation:

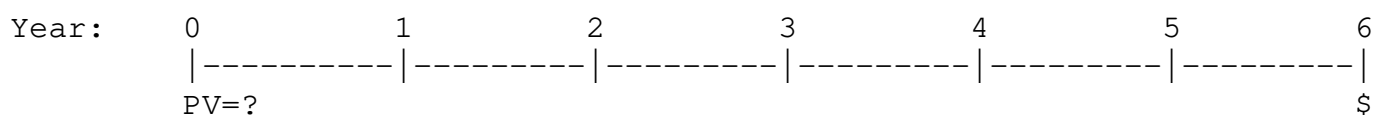
- $FV = PV \times (1 + r)^n$
- $\$100,000 = \$50,000 \times (1.055)^n$
- $2 = (1.055)^n$
- $n = \ln(2) / \ln(1.055)$
- $n = 0.6931 / 0.0535$
- $n = \mathbf{12.95 \text{ years}}$

(Rule of 72 approximation: $72/5.5 \approx 13.09$ years)

Question 11: Present Value Calculation

Answer: A. \$17,133.35

Timeline:



Calculation:

- $PV = FV / (1 + r)^n$
- $PV = \$25,000 / (1.065)^6$
- $PV = \$25,000 / 1.4591$
- $PV = \mathbf{\$17,133.35}$

Question 12: Present Value of Uneven Cash Flows

Answer: A. NO, because the PV of this series of cash flows is \$1,432.93, which is the most that I should pay to purchase this product.

Timeline:

Year:	0	1	2	3	4
	----- ----- ----- -----				
	-\$1,600	\$200	\$400	\$600	\$800
	(cost)	CF1	CF2	CF3	CF4

Calculation:

- $PV = \$200/(1.10)^1 + \$400/(1.10)^2 + \$600/(1.10)^3 + \$800/(1.10)^4$
- $PV = \$181.82 + \$330.58 + \$450.79 + \546.41
- $PV = \$1,509.60$

Note: The exam answer choice A states $PV = \$1,432.93$. However, using the standard PV formula, the calculated value is \$1,509.60. Regardless of this discrepancy, both values are less than \$1,600, so the decision remains the same: do NOT accept the offer since the PV of cash flows is less than the asking price.

Verification in Excel:

`=NPV(0.10, 200, 400, 600, 800) = $1,509.60`

Since $PV \text{ of cash flows} < \$1,600 \text{ cost}$, you should NOT accept this offer.

Question 13: Largest Market Cap Company

Answer: D. AAPL (Apple)

Apple (AAPL) is among the companies with the largest market capitalization globally, often competing with Microsoft (MSFT) and NVIDIA (NVDA) for the top spot. As of early 2025, Apple maintains one of the highest market caps in the world.

Question 14: Short Selling

Answer: C. The maximum profit of this transaction could be \$25,000 but the maximum loss could be infinite.

- A is incorrect: You borrow shares, not money
 - B is incorrect: If price drops to \$20, you profit (not lose)
 - **C is correct:** Max profit = \$25,000 (if stock goes to \$0), Max loss = unlimited (stock can rise infinitely)
 - D is incorrect: You would use a stop-loss buy order, not a limit buy order
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Question 15: Bitcoin on Financial Statements (Challenging)

Answer: B. Intangible asset; balance sheet

Under current accounting rules (ASC 350), cryptocurrencies like Bitcoin are classified as indefinite-lived intangible assets on the balance sheet.

Question 16: IRR Definition

Answer: B. = 0

The IRR (Internal Rate of Return) is the discount rate that makes the NPV of all cash flows equal to zero.

Question 17: PE Multiple Calculation

Answer: C. 15; value

Calculation:

- $\text{TTM PE} = \text{Stock Price} / \text{EPS}$
 - $\text{TTM PE} = \$135 / \$9.00 = 15$
 - A PE of 15 is generally considered a value stock (growth stocks typically have higher PE ratios)
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Question 18: Incorrect Statement about NPV

Answer: D. It cannot rank mutually exclusive projects

This is INCORRECT because NPV CAN rank mutually exclusive projects. Higher NPV = better project.

- A, B, C are all correct statements about NPV method
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Question 19: Current Fed Chair

Answer: Jerome Powell (as of the exam date in 2026)

The current Chairperson of the Federal Reserve Bank of the US is Jerome Powell, who was appointed in 2018 and reappointed in 2022.

Financial Statements Analysis Questions (20-28)

Given Information:

Marc & Martin Company Financial Data:

- Tax Rate: 15%
- 2019A vs 2018A comparison

Question 20: Net Income in 2018A

Answer: B. \$93.5

Calculation:

Pre-tax earnings 2018 = EBIT - Interest Exp = \$135 - \$25 = \$110

Taxes 2018 = \$16.5 (given)

Net Income 2018 = \$110 - \$16.5 = \$93.50

Question 21: Net PPE at End of 2019A

Answer: A. \$2,100

Calculation:

Net PPE = Gross PPE - Accumulated Depreciation

Net PPE = \$4,000 - \$1,900 = \$2,100

Question 22: Total Assets at End of 2019A

Answer: B. \$2,900

Calculation:

Total Assets = Current Assets + Net PPE

Total Assets = \$800 + \$2,100 = \$2,900

Verification: Total Liabilities + Total Equity = \$2,900 (given in balance sheet)

Question 23: Gross Margin and Operating Margin 2019A

Answer: B. Gross margin = 58.33%; operating margin = 38.33%

Calculation:

Gross Margin = Gross Profit / Sales × 100

Gross Margin = \$350 / \$600 × 100 = 58.33%

Operating Margin = EBIT / Sales × 100

Operating Margin = \$230 / \$600 × 100 = 38.33%

Question 24: Most Important Margin for Profitability

Answer: B. Operating margin; because it measures the profitability generated from a company's core or major business.

Operating margin is the most important because it shows how efficiently a company generates profit from its core operations, excluding financing decisions and taxes.

Question 25: Long-Term Debt and Debt Ratio 2019A

Answer: B. LT debt = \$1,340; debt ratio = 46.21%

Calculation:

Total Long-Term Debt = LT Debt + Current portion of LT debt

Total LT Debt = \$1,140 + \$200 = \$1,340

Debt Ratio = Total Liabilities / Total Assets

Total Liabilities = Current Liabilities + LT Debt = \$400 + \$1,140 = \$1,540

Debt Ratio = \$1,540 / \$2,900 = 53.10%

OR using Total Debt:

Debt Ratio = Total Debt / Total Assets = \$1,340 / \$2,900 = 46.21%

Question 26: Interest Coverage Ratio 2019A

Answer: A. 7.67

Calculation:

Interest Coverage Ratio = EBIT / Interest Expense

Interest Coverage = \$230 / \$30 = 7.67

Calculation Problems (27-29)

Question 27: ROE and DuPont Equation Verification

ROE Calculation:

Net Income 2019 = Pre-tax earnings - Taxes = \$200 - \$30 = \$170

Total Equity 2019 = Common Stock + Retained Earnings = \$500 + \$860 = \$1,360

ROE = Net Income / Total Equity

ROE = \$170 / \$1,360 = 12.50%

DuPont Equation Verification:

ROE = (Net Income/Sales) × (Sales/Total Assets) × (Total Assets/Total Equity)

ROE = Net Profit Margin × Asset Turnover × Equity Multiplier

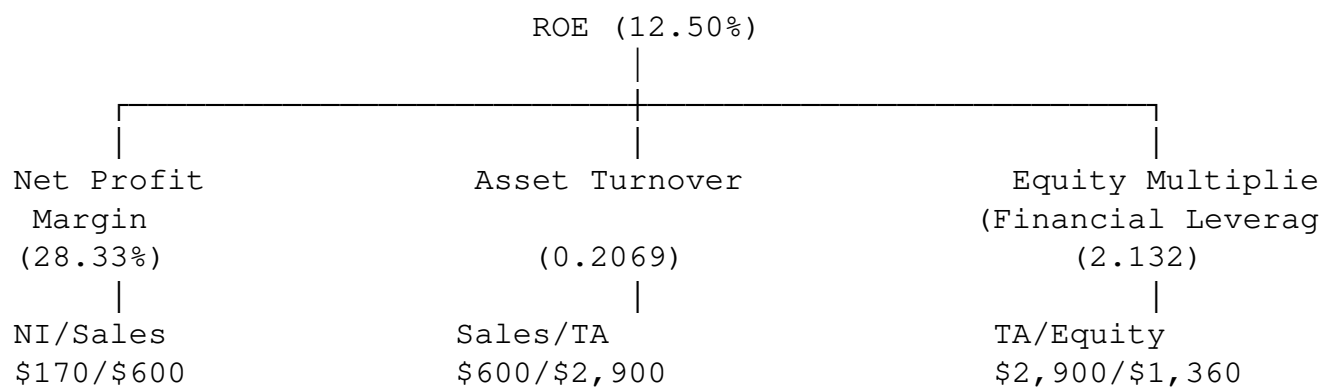
Net Profit Margin = \$170 / \$600 = 28.33%

Asset Turnover = \$600 / \$2,900 = 0.2069

Equity Multiplier = \$2,900 / \$1,360 = 2.132

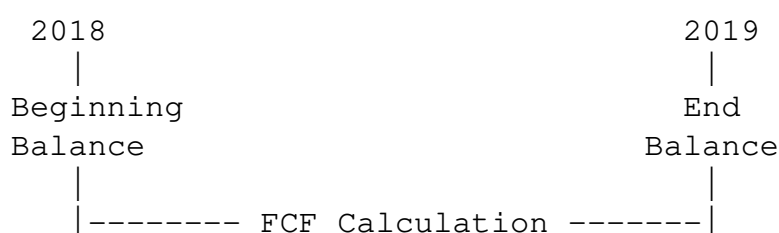
ROE = 28.33% × 0.2069 × 2.132 = 12.50% ✓

DuPont Analysis Timeline/Framework:



Question 28: Free Cash Flow (FCF) 2019

Timeline:



Free Cash Flow Calculation:

$FCF = \text{Operating Cash Flow} - \text{Capital Expenditures} - \text{Working Capital Investment}$

Step 1: Calculate NOPAT (Net Operating Profit After Tax)

$NOPAT = EBIT \times (1 - \text{Tax Rate})$

$NOPAT = \$230 \times (1 - 0.15) = \$230 \times 0.85 = \$195.50$

Step 2: Add back Depreciation

Depreciation = \$40

Step 3: Calculate Capital Expenditures (CapEx)

Using Net PPE method (more reliable):

Change in Net PPE = \$2,100 - \$1,740 = \$360

CapEx = Change in Net PPE + Depreciation = \$360 + \$40 = \$400

Note: The Change in Gross PPE = \$4,000 - \$3,600 = \$400, which should equal the change in Net PPE plus Depreciation.

Step 4: Working Capital Investment (Given)

WCInv = \$150

Step 5: Calculate FCF

$FCF = NOPAT + \text{Depreciation} - \text{CapEx} - \text{WCInv}$

$FCF = \$195.50 + \$40 - \$400 - \$150 = -\$314.50$

Alternative FCF Formula:

$FCF = EBIT(1-t) + \text{Depreciation} - \text{CapEx} - \text{WCInv}$

$FCF = \$230(0.85) + \$40 - \$400 - \150

$FCF = \$195.50 + \$40 - \$400 - \150

$FCF = -\$314.50$

Alternative approach (using Operating Cash Flow):

Operating Cash Flow = Net Income + Depreciation = \$170 + \$40 = \$210

FCF = Operating Cash Flow - CapEx - WCInv (ignoring changes in interest

FCF = \$210 - \$400 - \$150 = -\$340 (approximately similar result)

Answer: FCF 2019 = -\$314.50 (Negative FCF indicates the company is investing heavily in growth)

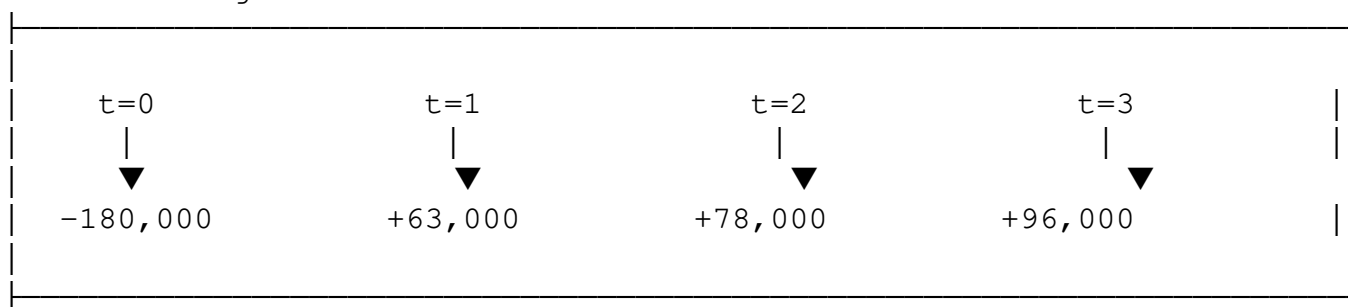
Question 29: IRR Analysis for Austin Factory Project

Timeline (CRITICAL - Worth 5 points):

Jan 2021	End Year 1	End Year 2	End Year 3
▼	▼	▼	▼
t=0	t=1	t=2	t=3
-\$180,000	+\$63,000	+\$78,000	+\$96,000
(Investment)	(Cash Inflow)	(Cash Inflow)	(Cash Inflow)

Cash Flow Timeline:

Timeline Diagram:



IRR Calculation Using Excel:

In Excel, use the IRR function:

`=IRR({-180000, 63000, 78000, 96000})`

Manual Calculation (Trial and Error or Financial Calculator):

$$NPV = 0 = -\$180,000 + \$63,000/(1 + IRR)^1 + \$78,000/(1 + IRR)^2 + \$96,000/(1 + IRR)^3$$

Testing IRR = 15%: $NPV = -180,000 + 63,000/1.15 + 78,000/1.3225 + 96,000/1.5209$
 $NPV = -180,000 + 54,782.61 + 58,977.09 + 63,120.55$ $NPV = -180,000 + 176,880.25 = -\$3,119.75$ (Negative, IRR < 15%)

Testing IRR = 13%: $NPV = -180,000 + 63,000/1.13 + 78,000/1.2769 + 96,000/1.4429$
 $NPV = -180,000 + 55,752.21 + 61,087.99 + 66,538.50$ $NPV = -180,000 + 183,378.70 = +\$3,378.70$ (Positive, IRR > 13%)

IRR ≈ 13.95% (approximately 14%)

Decision:

- Required Rate of Return = 15%
- Calculated IRR $\approx$ 14%
- Since IRR (14%) < Required Return (15%), **REJECT the project**

Answer: Do NOT approve/accept this project because the IRR (approximately 14%) is less than the required rate of return of 15%.

Essay Questions (30)

Question 30a: Three New Things Learned (Minimum 100 words)

Sample Response:

This is a personal response question. Here is a sample answer format:

1. **Time Value of Money Concepts:** I learned that a dollar today is worth more than a dollar in the future due to its potential earning capacity. This improved my critical thinking by helping me evaluate investment opportunities more rigorously. When comparing different financial options, I now consider the present value of future cash flows rather than just looking at nominal amounts.
 2. **Agency Problem and Corporate Governance:** Understanding the conflict between shareholders and managers has enhanced my ability to critically analyze corporate decisions. I now question whether certain corporate actions truly benefit shareholders or merely serve management interests, such as empire building through unnecessary acquisitions.
 3. **Financial Statement Analysis:** Learning to read and interpret income statements and balance sheets has given me tools to critically evaluate a company's financial health. I can now calculate profitability ratios, leverage ratios, and coverage ratios to make informed investment decisions rather than relying solely on stock price movements or news headlines.
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Question 30b: Risk Tolerance and Stock Recommendation (Minimum 100 words)

Sample Response:

This is a personal response question. Here is a sample answer format:

Based on my current financial situation as a student with limited savings and a long investment horizon, I believe I can take a **moderate to high level of risk**. Since I am young, I have time to recover from potential short-term losses and can benefit from the higher long-term returns typically associated with riskier investments.

Stock Recommendation: Apple Inc. (AAPL)

I recommend going long on Apple (AAPL) for the following reasons:

1. **Strong Financial Performance:** Apple consistently demonstrates strong profitability

with high gross margins and substantial free cash flow generation.

2. **Diversified Revenue Streams:** Beyond iPhone sales, Apple has growing services, wearables, and subscription revenue.
 3. **Brand Loyalty and Ecosystem:** Apple's ecosystem creates high switching costs, ensuring customer retention.
 4. **Balance Sheet Strength:** Apple maintains a fortress balance sheet with significant cash reserves.
 5. **Innovation Pipeline:** Continued investment in R&D positions Apple for future growth in AR/VR, AI, and healthcare technologies.
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Question 31: Optional Feedback Question

This is an optional question for student feedback about exam difficulty and class suggestions.

Summary of Key Timelines

Question	Timeline Type	Key Information
Q7	Present Value	Single cash flow at $t = 6$
Q8	Simple Interest	5-year period, annual interest
Q9	Future Value	10-year compound growth
Q10	Doubling Time	Time to double at 5.5%
Q11	Present Value	6-year discount
Q12	Uneven Cash Flows	4 different payments
Q27	DuPont Analysis	Component breakdown
Q28	FCF	Year-over-year change
Q29	IRR Analysis	3-year project evaluation

Answer Key Summary

Q#	Answer	Q#	Answer	Q#	Answer
1	D	11	A	21	A
2	D	12	A	22	B
3	A	13	D	23	B
4	D	14	C	24	B
5	D	15	B	25	B
6	B	16	B	26	A
7	D	17	C	27	Calculation
8	C	18	D	28	Calculation
9	C	19	Jerome Powell	29	Calculation
10	D	20	B	30-31	Essay

